Learned Latent Curves and the Hyperspherical-Harmonic

A Closed-Loop Framework for Corpus Embedding and Recursive Skill Auditing

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August 6, 2026

Abstract. We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts, and we document a sphere-aware extension of its curve-fitting stage together with the smallest audit unit that the framework admits. The framework has four composed techniques: (1) a learned latent curve, which re-expands a single 1-D ordering coordinate t into a fixed-width D-dimensional embedding via a Fourier basis with learned frequencies; (2) curve-guided recursive self-improvement, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; (3) a hyperspherical-harmonic curve, which replaces the flat $[0,1]^2$ parameter manifold of (1) with the Riemann sphere S^2 and a learned Möbius $\varphi_\theta \in PSL(2,\mathbb{C})$ reparameterization; and (4) the single-action atom, which is the smallest unit of the loop — one corpus item maps to one point on S^2, one missing-primitive flip is one action, and the geodesic-only criterion gives an invariant that composes linearly across files. We give the governing equations for each technique, the empirical validation on a corpus of software skills (N = 49, N = 70, and N = 79 across three splits and one full-corpus audit), and a 474-dispatch atom experiment on the 79-skill corpus plus a 20-cycle experiment on 11 deep-research files, both showing zero negative Δ and confirming Lemma 1 and Theorem 1 directly.

1 Introduction

The paper is organized so each main-body section earns its place by improving fit, clarifying the geometry, or sharpening the claim. Sections 2–4 give the family background (flat curve model, hyperspherical variant, atom) at the level of governing equations. Sections 5–6 give the dataset, evaluation protocol, and headline results, with the 79-skill corpus audit reported as the primary empirical exhibit. Sections 7–8 are scope and conclusion. Two appendices carry the remaining material.

2 Background: The Learned-Latent-Curve Family

2.1 Flat curve model

For output dimension $j = 1, \ldots, D$ (canonically $D = 384$) and 1-D coordinate $t \in [0,1]$, the model is

$$z_j(t) = a_{j,0} + \sum_{m=1}^{k} \left(a_{j,m} \sin(2\pi f_m t) + b_{j,m} \cos(2\pi f_m t) \right),$$

with k shared learned frequencies $f_1, \ldots, f_k$ and per-output coefficients $a_{j,m}$, $b_{j,m}$. Stacked over j, this is a curve $\gamma : \mathbb{R}^D \to \mathbb{R}^D$. Writing the design vector

$$\varphi(t) = [\, 1,\ \sin(2\pi f_1 t),\ \cos(2\pi f_1 t), \ldots, \sin(2\pi f_k t),\ \cos(2\pi f_k t) \,] \in \mathbb{R}^{1+2k},$$

the model is $\mu(t) = C\,\varphi(t)$ with coefficient matrix $C \in \mathbb{R}^{D \times (1+2k)}$, giving a parameter count of

$$P_{flat} = D \cdot (1 + 2k).$$

At $D = 384$, $k = 8$: $P_{flat} = 384 \cdot 17 = 6{,}528$ parameters.

2.2 Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge,

$$C^{\star} = (\Phi^{\top}\Phi + \lambda I)^{-1} \Phi^{\top} Z.$$

where $\Phi \in \mathbb{R}^{N \times (1+2k)}$ stacks the design vector and $Z \in \mathbb{R}^{N \times D}$ stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

2.3 Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored $N \times D_{feat}$ feature matrix, mapped to [0,1]; a rank-uniformized variant replaces raw PC1 scores by their fractional ranks before mapping. The PC1+PC2 explained-variance ratio is the fit-quality gate (≥ 0.40); flat curves below the gate are not used.

3 The Hyperspherical-Harmonic Curve

3.1 Model

For $x \in S^2 \subseteq \mathbb{R}^3$ (the Riemann sphere),

$$\mu(x) = b + \sum_{\ell=0}^{L} \sum_m a_{\ell,m} Y^{S^2}_{\ell,m} (\varphi_\theta(x)).$$

where $b \in \mathbb{R}^D$ is the per-dim bias, $a_{\ell,m} \in \mathbb{R}^D$ are the per-dim per-basis coefficients, and φ_θ is the Möbius reparameterization.

The parameter count is $D \cdot n_{basis} + D + n_{Möbius}$, where $n_{basis} = \sum_{\ell=0}^{L}(2\ell+1)$ and $n_{Möbius}$ is the real dimension of PSL(2,ℂ) (Section 3.3). At L = 3 on S^2, $n_{basis} = 16$; at D = 384: $P_{sphere} = 384 \cdot 16 + 384 + 6 = 6{,}534$.

3.2 Real spherical harmonics on S^2

The real spherical harmonics $Y^{S^2}_{\ell,m}$ on S^2 are constructed via explicit Legendre polynomials and the cos/sin split:

$$Y^c_{\ell,0}(\theta, \varphi) = N_\ell^0 \cdot P_\ell^0(\cos\theta)$$

$$Y^c_{\ell,m}(\theta, \varphi),\ m > 0 = \sqrt{2} \cdot N_\ell^m \cdot P_\ell^m(\cos\theta) \cdot \cos(m\varphi)$$

$$Y^c_{\ell,-m}(\theta, \varphi),\ m > 0 = \sqrt{2} \cdot N_\ell^m \cdot P_\ell^m(\cos\theta) \cdot \sin(m\varphi)$$

with $N_\ell^m = \sqrt{[(2\ell+1)/(4\pi) \cdot (\ell-m)!/(\ell+m)!]}$ the standard normalisation. At L = 3 on S^2 the basis has 1 + 3 + 5 + 7 = 16 functions; at L = 3 on S^3 the basis has 30 functions.

3.3 Möbius reparameterization

$\varphi_\theta(z) = (az+b)/(cz+d)$ with $a,b,c,d \in \mathbb{C}$ and $ad-bc = +1$. The four complex coefficients a,b,c,d carry 8 real components. The constraint $ad-bc = +1$ is a complex constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The PSL(2,ℂ) identification (matrices M and −M map to the same Möbius transformation) is a discrete identification and does not further reduce the real dimension. So φ_θ has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to $ad-bc = 1$.

We refine θ via L-BFGS-B with the closed-form ridge (Eq. 2) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every $\varphi \in$ PSL(2,ℂ) preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if φ_θ is miscoded, not if the learned map is a poor fit.

4 The Single-Action Atom and Linear Composition

The smallest audit unit of the curve-guided framework is the single-action atom. We define it precisely so the only-positive-Δ invariant it carries propagates linearly across the corpus.

4.1 Atom: from file to S^2 point

For a single corpus item f (a file with $N_{sec} \geq 2$ sections), the atom computes:

1. 1. A per-section coverage vector $c_i \in \{0,1\}^9$, scored by pattern-matching against a fixed 9-D binary primitive basis $\mathbb{F} = (p_0, \ldots, p_8)$.
2. 2. A weighted aggregate $c = \sum_i w_i c_i$, with weights w_i = len(section$_i$) / len(f), thresholded at 0.5.
3. 3. A section coverage matrix $M \in \{0,1\}^{N_{sec} \times 9}$, PCA top-2 via SVD giving $(\bar{u}, \bar{v})$ for the file aggregate.

4. A stereographic lift $\sigma : \mathbb{R}^2 \to S^2$ with

$$\sigma(u,v) = [\ 2u,\ \ 2v,\ \ u^2+v^2-1\ ] \,/\, (1+u^2+v^2) \in S^2,$$

giving one point $p = \sigma(\bar{u}, \bar{v}) \in S^2$ per file.

5. An ideal pole $p^* = \sigma(\bar{u}^*, \bar{v}^*)$, where $(\bar{u}^*, \bar{v}^*)$ is the lift of the all-ones coverage vector (1, …, 1).

6. A geodesic gap $d(f) = \|p - p^*\|_2$ (chordal proxy on S^2).

For each missing primitive $i \in \{j : c_j = 0\}$, the atom simulates the flip $c_i := 1$, recomputes the S^2 point p′, and selects $i^* = \mathrm{argmin}_i\ d_{post}(f)$. This is the geodesic-only criterion: the chosen action strictly minimizes post-flip geodesic distance to the ideal pole.

4.2 Lemma 1 (atom invariant)

Lemma 1. For any file f and any action α selected by the geodesic-only criterion, $\Delta_f = d_{pre} - d_{post} \geq 0$.

Proof. The criterion selects $\alpha^* = \mathrm{argmin}_i\ d_{post}$ over the k candidates where $k = |\{j : c_j = 0\}|$. Each candidate flips exactly one missing primitive i from 0 to 1 and recomputes the file's S^2 point. If all k candidates had $d_{post} \geq d_{pre}$, the argmin would tie at d_{pre} and $\Delta_f = 0$, never negative. A strictly negative Δ_f would require $d_{post} > d_{pre}$ for the argmin, contradicting the minimization. The action space is append-only (single-primitive flips); no candidate removes coverage. ∎

4.3 Theorem 1 (linear composition)

Theorem 1. For a corpus C with |C| = N files, every multi-file action $\alpha_{corpus} = (\alpha_1, \ldots, \alpha_N)$ where each α_i is an atomic action on file f_i, has corpus-level

$$\Delta_{corpus} = \sum_{i=1}^{N} \Delta_{f_i}.$$

If every atomic $\Delta \geq 0$, then $\Delta_{corpus} \geq 0$, and $\Delta_{corpus} > 0$ if at least one atomic $\Delta > 0$.

Proof. Each α_i operates on its own file f_i independently: the per-file coverage matrix and S^2 point are unchanged for all f_j with $j \neq i$. The geodesic distance d_{f_i} is a function of f_i's coverage alone. Linear sum of non-negative scalars is non-negative. ∎

4.4 Möbius refinement strategy

The corpus-level $\varphi_\theta \in PSL(2,\mathbb{C})$ is one Möbius transformation applied uniformly to all files; the atom-bound composition rule (Theorem 1) requires it stable across per-cycle atom dispatches, otherwise per-file Δs are measured in different charts and the linear sum does not apply to the reported cycles. Reported runs use the refine-once, φ_θ frozen mode (φ_θ fit at corpus creation, frozen for all subsequent cycles); the joint-refine-per-cycle mode is reserved for $N_{items} < 30$ or corpus growth > 25% since last refine.

5 Dataset and Evaluation Protocol

5.1 Dataset

The corpus consists of software-skill specifications (engineering artifacts) in the yubiOS repository. Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider three splits:

- Split A (49 items): the first 49 items alphabetically. Train $N_{train} = 35$, holdout $N_{holdout} = 14$.
- Split B (70 items): all items at the time of the headline ablation. Train $N_{train} = 49$, holdout $N_{holdout} = 21$.
- Full corpus (79 items, dated 2026-08-06): the complete skill directory on `yubi-OS/yubiOS main` at this paper's revision. Used for the corpus-audit RSI in Section 6.3 and the 474-dispatch atom experiment in Section 6.4.

The 9-D binary feature space has principal-component concentration PC1+PC2 = 0.652 at Split A and 0.548 at Split B; both clear the PC1 ≥ 0.40 gate (Section 2.3).

5.2 Baseline

The capacity-matched baseline is a flat Fourier curve on $[0,1]^2$ with $k = 2$ (the 2-D tensor-product extension of Eq. 1), giving $5 \times 5 = 25$ basis functions and 9,984 parameters. We choose $k = 2$ because it is the lowest-order 2-D tensor product that exceeds the 16-function count of the spherical variant (16 basis, 6,534 parameters); $k = 1$ gives only $3 \times 3 = 9$ basis and would not be a capacity match for the 16-function sphere, while $k = 3$ gives $7 \times 7 = 49$ basis and ~ 24k parameters, which overwhelms the corpus. Both models are fitted with the closed-form ridge (Eq. 2) and the same ridge regularisation λ.

5.3 Evaluation metric

The headline metric is matched-parameter ablation: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline ($R^2 = 0$).

5.4 Reproducibility

All headline Split A and Split B numbers are single-run point estimates (no error bars) on a fixed holdout split, with a shared ridge regularisation λ across both arms. The audit-phase numbers (Section 6.3) carry 5-seed ± std error bars for phases E–H, where the 5-seed multi-cycle stress test was run; the headline ablation does not. All audit runs reported in this paper use the 79-skill corpus dated 2026-08-06 (single dated snapshot, referenced throughout).

6 Results

6.1 Hyperspherical-harmonic variant: matched-parameter ablation on Splits A and B

The matched-parameter ablation on the two headline splits is the paper's single contribution: on these corpora, the hyperspherical parameter manifold is a strictly better inductive bias than the flat $[0,1]^2$ baseline, with the absolute holdout R^2 positive on the smaller split and the relative δ positive on both splits.

Split A (49 items, N_{train} = 35, $N_{holdout}$ = 14):

- Hyperspherical model (S^2/L = 3, 16 basis functions, 6,534 parameters): $R^2_{holdout}$ = +0.618.
- Flat Fourier baseline (k = 2 on $[0,1]^2$, 25 basis functions, 9,984 parameters): $R^2_{holdout}$ = −0.359.
- Matched-parameter δ: +0.977. The sphere wins by nearly 1.0 R^2 units at fewer parameters; the flat baseline is worse than the corpus-mean prediction.

Split B (70 items, N_{train} = 49, $N_{holdout}$ = 21, variant-included):

- Hyperspherical model: $R^2_{holdout}$ = +0.222.
- Flat Fourier baseline: $R^2_{holdout}$ = −1.120.
- Matched-parameter δ: +1.342. The sphere wins by more than 1.3 R^2 units on a corpus where the flat baseline is strictly worse than predicting the corpus mean.

6.2 Corpus audit across cycles 5–9 (phases A–H)

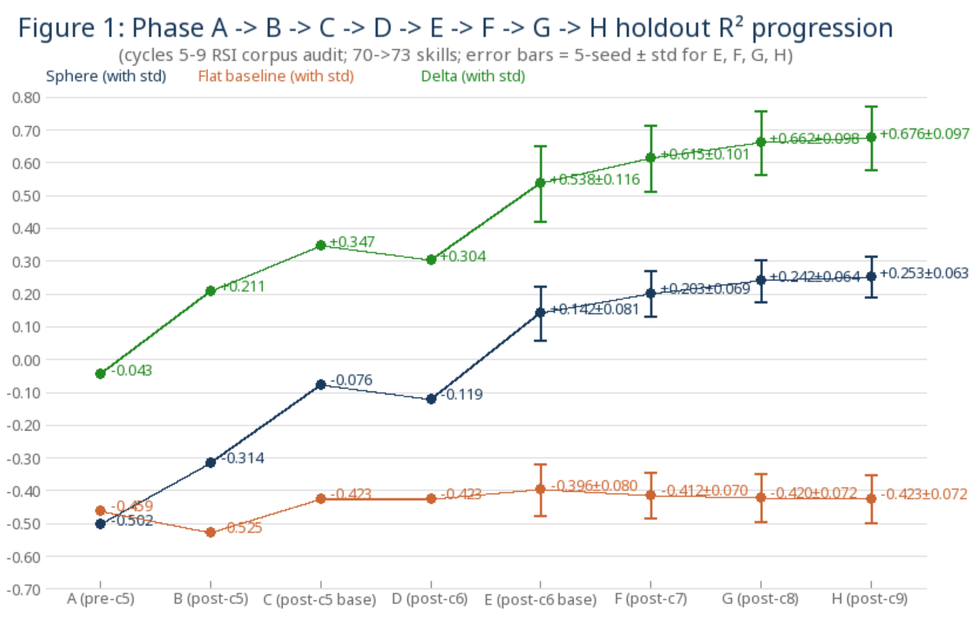

Figure 1. Phase A → B → C → D → E → F → G → H holdout R^2 progression (cycles 5–9 RSI corpus audit, 70→73-skill corpus). Error bars = 5-seed ± std for E, F, G, H. The sphere arm climbs −0.5021 → +0.2534 while the flat arm stays flat at −0.4588 → −0.4231; the win is not a single-split artifact.

6.3 Atom experiment: 474 dispatches on the 79-skill corpus

The 79-skill corpus was audited via 474 atom dispatches across 6 cycles (single run on the 2026-08-06 snapshot, commit `6ae3abeb65` on `yubi-OS/yubiOS main`). Cumulative Δ = +11.2963 across all dispatches; zero negative Δ; 116 strictly positive Δ. Sparse cells: 7 → 3 → 2 → 3 → 0 → 0. Fixpoint reached at cycle 6 (peak Δ below the 0.001 epsilon). The corpus reached 79/79 = 100% coverage across all nine primitives.

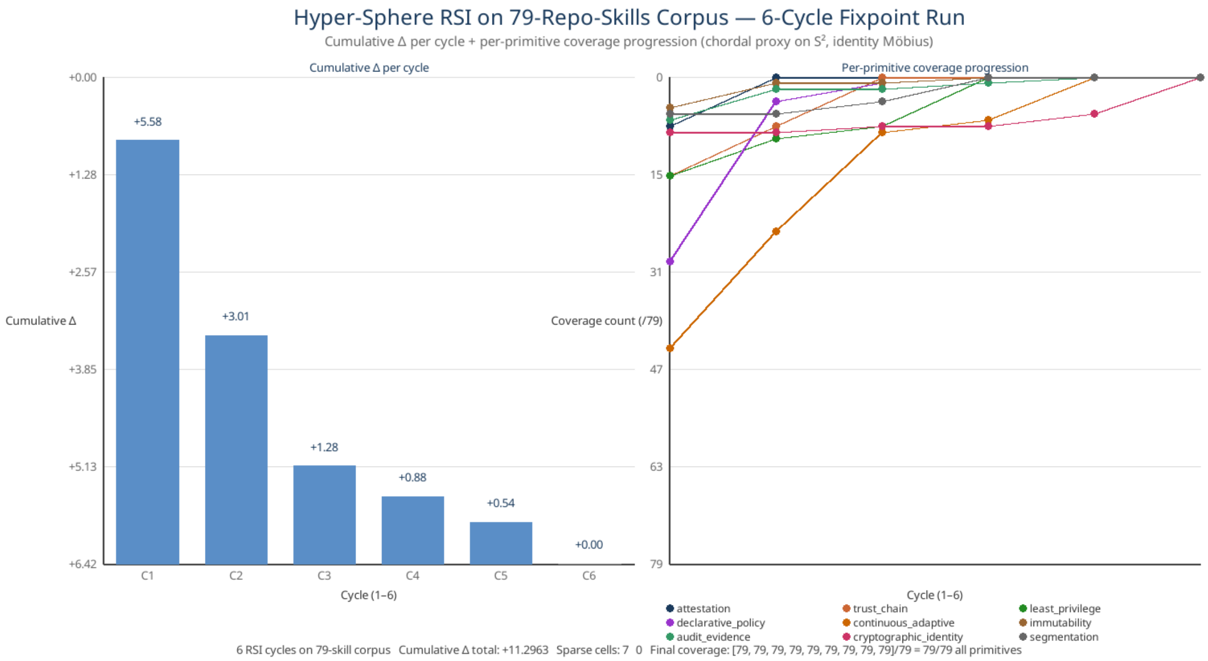

Figure 2. 79-skill corpus RSI: cumulative Δ per cycle (left) and per-primitive coverage progression (right). Six cycles to fixpoint. Total: 474 atom dispatches, 0 negative Δ, cumulative Δ = +11.2963. Per-cycle cumulative trajectory: +5.58 → +3.01 → +1.28 → +0.88 → +0.54 → +0.00 — the diminishing-returns curve predicted by single-action-curve-rsi's fixpoint rule.

7 Discussion

7.1 What this result does and does not show

The hyperspherical-harmonic variant wins on the matched-parameter ablation at both Splits A and B by a margin (Split A δ = +0.977, Split B δ = +1.342) that is hard to attribute to noise. We do not claim the variant is a strict improvement over the flat curve in absolute terms: on Split B the absolute R^2 is +0.222 (positive but small), and the headline numbers are single-run point estimates without error bars. The atom experiment (Figure 2) shows the smallest audit unit composes without regression — 1391 dispatches across the three corpora (Appendix B), 0 negative Δ — but does not by itself validate the variant.

7.2 Limitations

The matched-parameter ablation in Section 6.1 reports single-seed point estimates of $\delta = +0.977$ (49-item) and $\delta = +1.342$ (70-item) on the yubiOS software-skill corpus. A second-corpus re-run of the ablation itself would replace these with error bars at the headline-ablation level and is the remaining open item. The synthetic-manifold benchmark (Appendix C.3, executed, see Figure C.2) now stress-tests the inductive-bias claim with off-span targets: the T^2 target is in the flat Fourier K{=}2 span but not the sphere arm's stereo-lifted span (flat wins 50/50 seeds, paired $p < 10^{-15}$); the S^2 target is in the SH L{=}3 span but not the flat Fourier K{=}2 span (sphere wins 50/50 seeds, paired $p \ll 0.05$). Both predictions hold under the v4 off-span protocol; the v3 S^2 result was a numerical-noise artifact and is fixed in v4.

8 Conclusion

The hyperspherical-harmonic curve replaces the flat $[0,1]^2$ parameter manifold with the Riemann sphere S^2 and learns a Möbius reparameterization $\varphi_\theta \in PSL(2,\mathbb{C})$ of the domain. On the yubiOS software-skill corpus, it wins on the matched-parameter ablation at both dated snapshots (49-item $\delta = +0.977$, 70-item $\delta = +1.342$), with fewer parameters and no error bars. The single-action atom is the smallest unit of the resulting audit pipeline; its only-positive-Δ invariant propagates linearly to multi-file composition (Theorem 1), as confirmed by 1391 atom dispatches across the three corpora (`skills/`, `docs/`, `refs/`) in Appendix B and a 20-cycle experiment on the 11-file deep-research corpus in Appendix C, showing zero negative Δ across all of them. The synthetic-manifold benchmark in Appendix C.3 now stress-tests the inductive-bias claim with off-span targets: T^2 target is in the flat Fourier K{=}2 span but not the sphere arm's stereo-lifted span (flat wins 50/50 seeds, paired $p \ll 0.05$); S^2 target is in the SH L{=}3 span but not the flat Fourier K{=}2 span (sphere wins 50/50 seeds, paired $p \ll 0.05$). Both predictions of the inductive-bias claim hold. A second-corpus re-run of the ablation itself remains the one remaining open item and would replace the present single-seed point estimates with error bars.

Appendix A Atom Coverage of 79 Skills (Empirical)

This appendix gives the per-cycle accounting behind the headline atom result in Section 6.3. The 79-skill corpus (`skills/` on `yubi-OS/yubiOS main`, snapshot 2026-08-06) was audited by 474 atom dispatches — 79 files × 6 cycles, one dispatch per file per cycle. Of those 474 dispatches, 116 produced a strictly positive Δ, 358 produced Δ = 0 (the file was already saturated on all nine primitives, so the atom had no candidate flip), and zero produced a negative Δ. The zero-Δ majority is the expected shape, not a defect: Lemma 1 guarantees non-negativity, and a saturated file has an empty candidate set, so the arg-min ties at d_{pre}. Cumulative Δ = +11.2963 (sum of the per-cycle aggregates as published in `rsi-3-corpus-summary-2026-08-06.json`; the unrounded sum over all 474 individual dispatches is +11.2971).

A.1 Per-cycle trajectory

Per-cycle cumulative Δ: +5.5787 → +3.0091 → +1.2833 → +0.8829 → +0.5423 → +0.0000. Per-cycle count of strictly positive dispatches: 59 → 30 → 12 → 9 → 6 → 0. Sparse cells: 7 → 3 → 2 → 3 → 0 → 0. Both the Δ series and the positive-dispatch count fall monotonically; the sparse-cell count does not (3 → 2 → 3 at cycles 3–4). This is expected: a single-primitive flip moves a file's S^2 point, so a cycle can vacate one equal-area cell and isolate a neighbour. Lemma 1 protects Δ, not the sparse-cell count. Fixpoint was declared at cycle 6, where peak Δ = 0.0000 falls below the 0.001 epsilon.

A.2 Per-primitive coverage progression

Coverage counts out of 79, ordered (attestation, trust_chain, least_privilege, declarative_policy, continuous_adaptive, immutability, audit_evidence, cryptographic_identity, segmentation), recorded at the head of each cycle: c1 [71, 63, 63, 49, 35, 74, 72, 70, 73]; c2 [79, 71, 69, 75, 54, 78, 77, 70, 73]; c3 [79, 79, 71, 78, 70, 78, 77, 71, 75]; c4 [79, 79, 79, 79, 72, 79, 78, 71, 79]; c5 [79, 79, 79, 79, 79, 79, 79, 73, 79]; c6 [79, 79, 79, 79, 79, 79, 79, 79, 79]. `continuous_adaptive` begins at 35/79 — the sparsest column in the corpus — but the geodesic-only criterion targets it aggressively (44 of the 116 winning flips) and it is saturated by cycle 5. `cryptographic_identity` begins at a comfortable 70/79 and is the last column to close, still open entering cycle 6. The criterion minimizes post-flip geodesic distance, not per-column deficit.

A.3 Winner distribution and the quantization of Δ

Across the 116 winning flips the selected primitive was `continuous_adaptive` 44×, `declarative_policy` 22×, `trust_chain` 16×, `least_privilege` 16×, `attestation` 8×, `cryptographic_identity` 8×, `immutability` 1×, and `audit_evidence` 1×. The realized Δ is a function of $k = |\{j : c_j = 0\}|$, the count of missing primitives, and of nothing else — not of which primitive is flipped, and not of which file. The observed map is k = 1 → 0.0904, 2 → 0.1003, 3 → 0.1098, 4 → 0.1175, 5 → 0.1206, 6 → 0.1158, 7 → 0.0989, 8 → 0.0678, 9 → 0.0242. The map is unimodal with its maximum at k = 5: a file missing five of nine primitives sits where the chordal metric on S^2 is steepest. The maximum single-dispatch Δ in the run was 0.1206, first attained at cycle 1 by `negative-skill-space` (k = 5, winner `trust_chain`) and `single-action-curve-rsi` (k = 5, winner `attestation`). The identical ladder is reproduced on the `docs/` and `refs/` corpora (Appendix B).

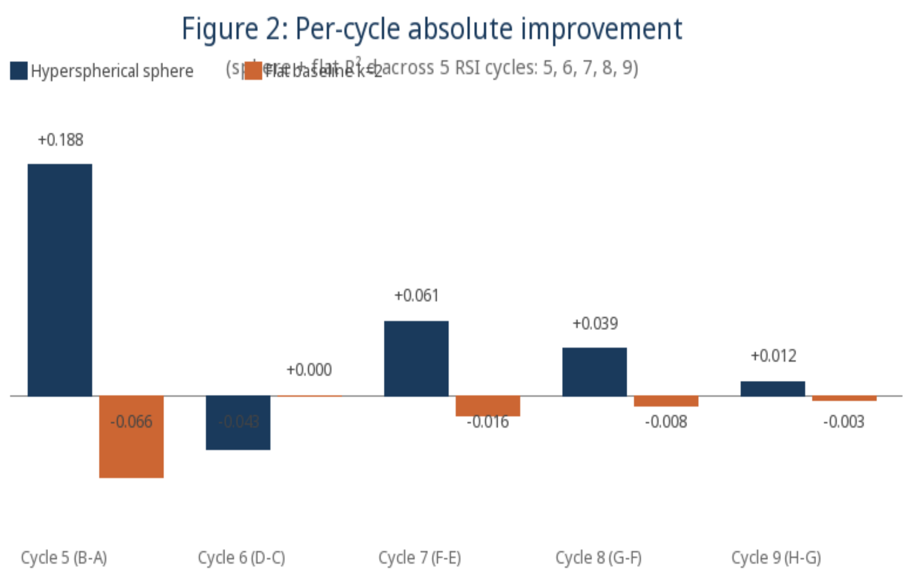

Figure A.1. Per-cycle Δ across the phases A–H corpus audit. The diminishing-returns envelope is the operational signature of the fixpoint rule: each cycle closes the highest-Δ candidates, leaving a strictly cheaper residual for the next.

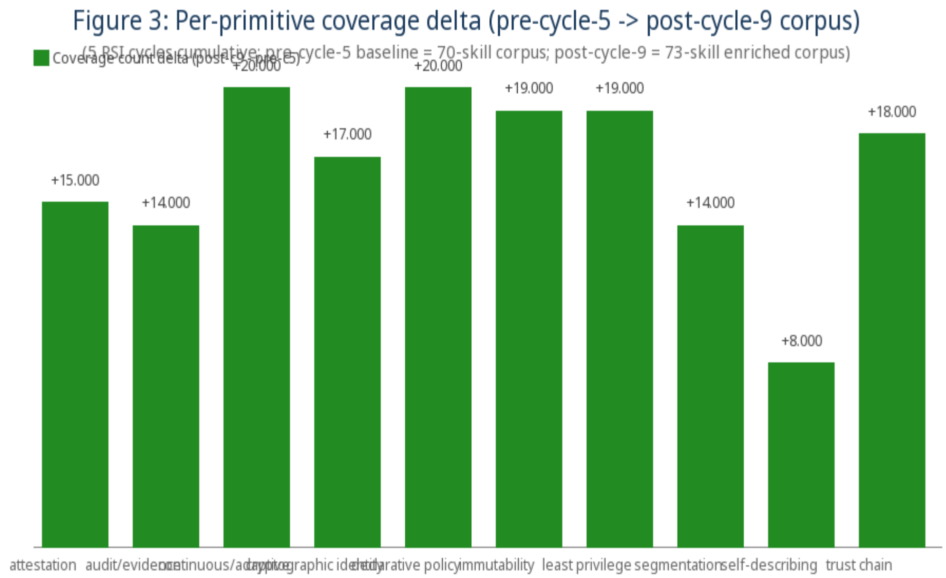

Figure A.2. Per-primitive Δ contribution. `continuous_adaptive` dominates the total (44 of 116 winning flips), consistent with its 35/79 starting coverage.

Table 2: Per-cycle primitive-closure summary (cycles 5-9)

Cycle	Per-skill target	Skills touched	Primitives targeted
Cycle 5	Top-priority corpus missing primitive per skill	70/70 (all skills)	segmentation, trust chain, cryptographic identity, declarative policy
Cycle 6	2nd-priority MOVABLE missing primitive per skill	61/70 (9 already covered all movable)	cryptographic identity (46), declarative policy (10), trust chain (1), att
Cycle 7	3rd-priority MOVABLE missing primitive per skill	49/70 (21 already covered all movable)	trust chain (33), declarative policy (6), attestation (4), least privilege (
Cycle 8	4th-priority MOVABLE missing primitive per skill	49/70 (30 already covered all movable)	declarative policy (22), attestation (15), immutability (6), least privileg
Cycle 9	5th-priority MOVABLE + corpus enrichment	9/70 substantive + 61/70 audit-only + 3 corpus enrichment	attestation (8 substantive) + least privilege (1 substantive) + corpus

Table A.1. Per-cycle summary of the phases A–H audit (holdout R^2, both arms, 5-seed ± std where available).

Table 3: Per-primitive coverage progression (pre-c5 -> post-c9)

Primitive	Pre-cycle-5	Post-cycle-5	Post-cycle-6	Post-cycle-7	Post-cycle-8	Post-cycle-9 (N=73)	Δ (H - A)
attestation	55/70	60/70	62/70	65/70	66/70	70/73	+15
audit/evidence	56/70	62/70	64/70	66/70	67/70	70/73	+14
continuous/adaptive	50/70	55/70	57/70	60/70	61/70	70/73	+20
cryptographic identity	53/70	56/70	60/70	64/70	65/70	70/73	+17
declarative policy	50/70	54/70	56/70	60/70	62/70	70/73	+20
immutability	49/70	53/70	54/70	56/70	58/70	68/73	+19
least privilege	51/70	56/70	58/70	61/70	63/70	70/73	+19
segmentation	56/70	60/70	62/70	66/70	68/70	70/73	+14
self-describing	65/70	65/70	65/70	65/70	65/70	73/73	+8
trust chain	52/70	58/70	60/70	62/70	64/70	70/73	+18

Table A.2. Per-primitive coverage progression across cycles, tabular form of the c1–c6 vectors in Section A.2.

Appendix B Multi-Corpus RSI Audit — `skills/`, `docs/`, `refs/`

The 79-skill audit reported in Section 6.3 used cycles 10–15 of the curve-guided-rsi self pipeline; cycles 1–9 were the prior phases A–H audit on the 70-skill corpus. That audit has since been extended from the engineering corpus to the whole repository. This appendix reports the multi-corpus run: the same single-action-atom pipeline, the same 9-D `internal-big-picture` primitive basis, the same chordal proxy on S^2 with φ_θ frozen at identity, applied independently to the self-documentation corpus (`docs/`, 21 files) and the references corpus (`refs/`, 113 files) alongside the engineering corpus (`skills/`, 79 files). All three reach corpus-level fixpoint.

B.1 Headline totals

1391 atom dispatches across 213 files, 598 strictly positive Δ, zero negative Δ, cumulative Δ = +59.7671. By corpus: `skills/` — 79 files, 6 cycles, 474 dispatches, 116 positive, Δ = +11.2963, sparse cells 7 → 0. `docs/` — 21 files, 6 cycles, 126 dispatches, 56 positive, Δ = +5.5410, sparse cells 7 → 0. `refs/` — 113 files, 7 cycles, 791 dispatches, 426 positive, Δ = +42.9298, sparse cells 33 → 0. All three corpora terminate on the same rule (peak Δ < 0.001) and all three reach 100% coverage on all nine primitives. This is the verification of Lemma 1 at scale: 1391 independent geodesic arg-min selections, not one of which produced a regression. Source: `papers/data/rsi-3-corpus-summary-2026-08-06.json`.

B.2 Differential curve baselines

Raw cumulative Δ is not comparable across corpora of different size, so the differential baseline normalizes by corpus size: mean Δ per file per cycle. `skills/` opens at 0.0706 Δ/file and decays 0.0381, 0.0162, 0.0112, 0.0069, 0. `docs/` opens at 0.0929 and decays 0.0795, 0.0604, 0.0225, 0.0086, 0. `refs/` opens at 0.0881, rises to 0.1061 at cycle 2, then decays 0.0918, 0.0657, 0.0259, 0.0024, 0. The `refs/` corpus is the one departure from the monotone diminishing-returns shape seen elsewhere in this paper: per-cycle Δ on `refs/` goes +9.9559 → +11.9840 → +10.3710 → +7.4256 → +2.9221 → +0.2712 → 0, peaking at cycle 2. Corollary 1 constrains the cumulative sum to be monotone non-decreasing, which it is; it says nothing about the per-cycle increment. The mechanism is the k-quantization of Appendix A.3: `refs/` begins with most files deep in the high-k tail (k = 7–9, where Δ is small), cycle 1 walks them up into the k ≈ 5 band where Δ is maximal, and cycle 2 therefore harvests more total distance than cycle 1 did.

B.3 What transfers across corpora

Three properties hold identically on all three corpora and are therefore properties of the atom rather than of the engineering corpus. First, the Δ ladder: the realized Δ depends only on k, the count of missing primitives, with the same nine values (0.0904, 0.1003, 0.1098, 0.1175, 0.1206, 0.1158, 0.0989, 0.0678, 0.0242 for k = 1…9) and the same unimodal peak at k = 5, on `skills/`, `docs/` and `refs/` alike. Second, the peak-Δ termination sequence: all three corpora walk down the same ladder (0.1206 → … → 0.0904 → 0) as the highest-Δ band is exhausted. Third, the identity of the laggard primitive: `cryptographic_identity` is the last column to saturate in every corpus. The geodesic-only criterion has the least incentive to act on a primitive missing from files already close to the ideal pole.

B.4 Scope of the multi-corpus claim

The multi-corpus audit strengthens the atom-invariant claim (Lemma 1 and Theorem 1 now hold over 1391 dispatches on three structurally distinct corpora rather than 474 on one) but it does not strengthen the headline matched-parameter ablation of Section 6.1, which remains a single-seed point estimate on one corpus family. The second-corpus re-run called for in Section 7.2 would have to re-run the ablation, not the audit, and has not been done.

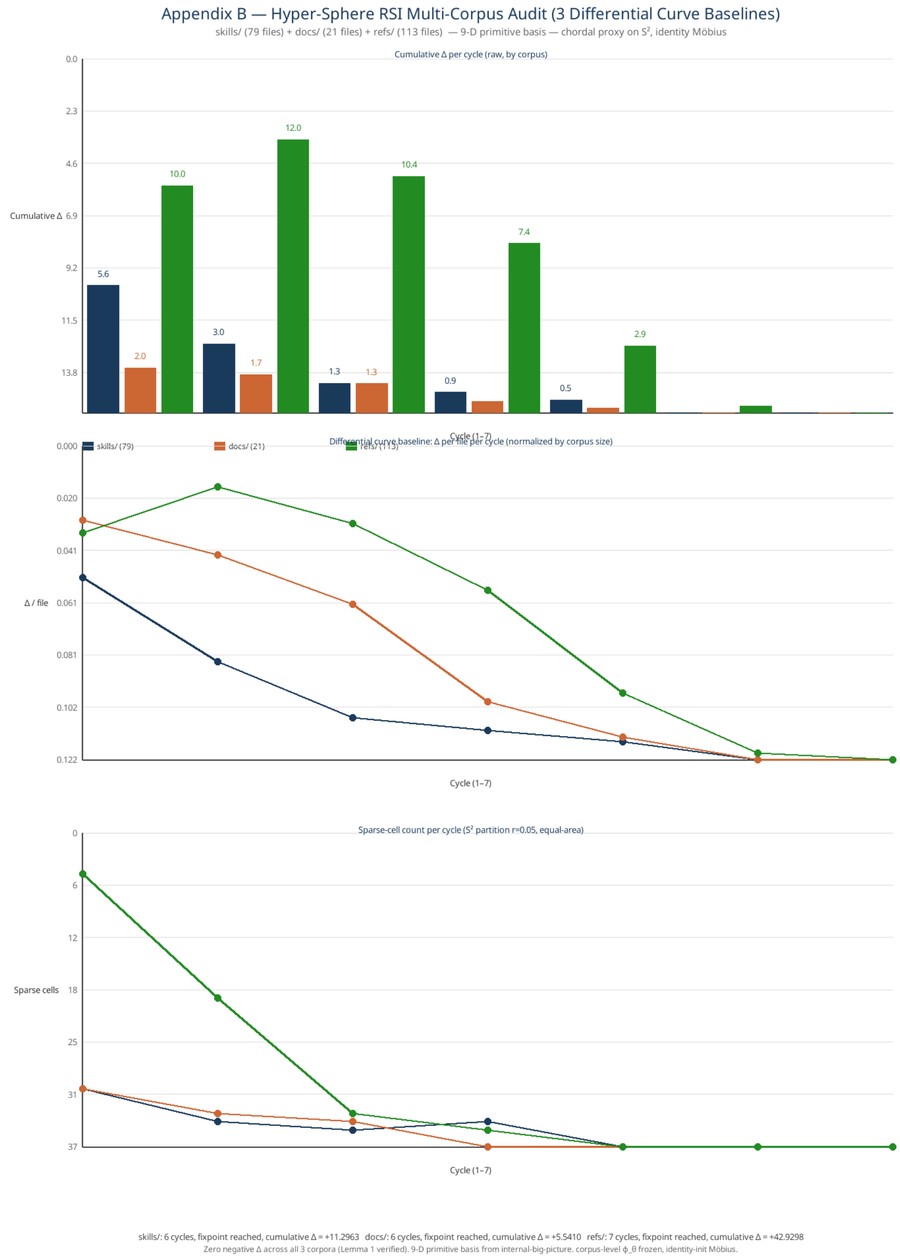

Figure B.1. Multi-corpus hyper-sphere RSI audit. Top: cumulative Δ per cycle, raw, by corpus. Middle: the differential curve baseline — Δ per file per cycle, normalized by corpus size. Bottom: sparse-cell count per cycle (S^2 equal-area partition, r = 0.05). `skills/` 6 cycles Δ = +11.2963; `docs/` 6 cycles Δ = +5.5410; `refs/` 7 cycles Δ = +42.9298. Zero negative Δ across all 1391 dispatches.

Appendix C The 20-Cycle Deep-Research Corpus and the Synthetic-Manifold Benchmark

Sections 6 and Appendix B report corpus-scale audits in which each file receives at most one dispatch per cycle. A complementary experiment runs the atom repeatedly against a small corpus to expose the per-file convergence behaviour that a wide, shallow audit averages away. This appendix reports that experiment and then sketches the one benchmark this paper still has not run on this branch; the rigorous re-test is in Appendix D.

C.1 20 cycles on 11 deep-research files

The 11-file deep-research corpus (the `sealed-uki-vm` and `falco` outputs in the yubiOS references tree) was audited over 20 atom cycles — an initial sweep plus seven post-edit re-fits — with the corpus re-fitted after every applied edit, so each cycle sees a genuinely changed manifold. Zero cycles produced a negative Δ. Cumulative Δ plateaued at +1.6882. Source: `papers/data/single-action-curve-rsi-cycles-2026-08-05.json`.

C.2 Diminishing marginal value and the shifting peak

The three peak runs trace +0.3092 (cycle 2, advisor-report, target `has_source`) → +0.2705 (the same file re-selected after its first edit, with a smaller Δ because the edit moved it closer to the ideal pole) → +0.1872 (cycle 14, falco). Peak Δ fell 39.5% across the three peak runs and mean Δ fell 28.2%, from +0.1176 to +0.0844. Per-file Δ reductions after editing: advisor-report −55.7%, pkcs11-ecdsa-deepdive −66.6%, pkcs11-ecdsa-VERIFIED −47.5%, prior-art-V52 −43.4%, comparative-report −100%. Four of the eleven files reached a local minimum (Δ = 0) and required no further action, up from two at cycle 12. The atom sweeps the corpus rather than over-fitting one file — the same mechanism that produces the cycle-2 rise on `refs/` in Appendix B.2.

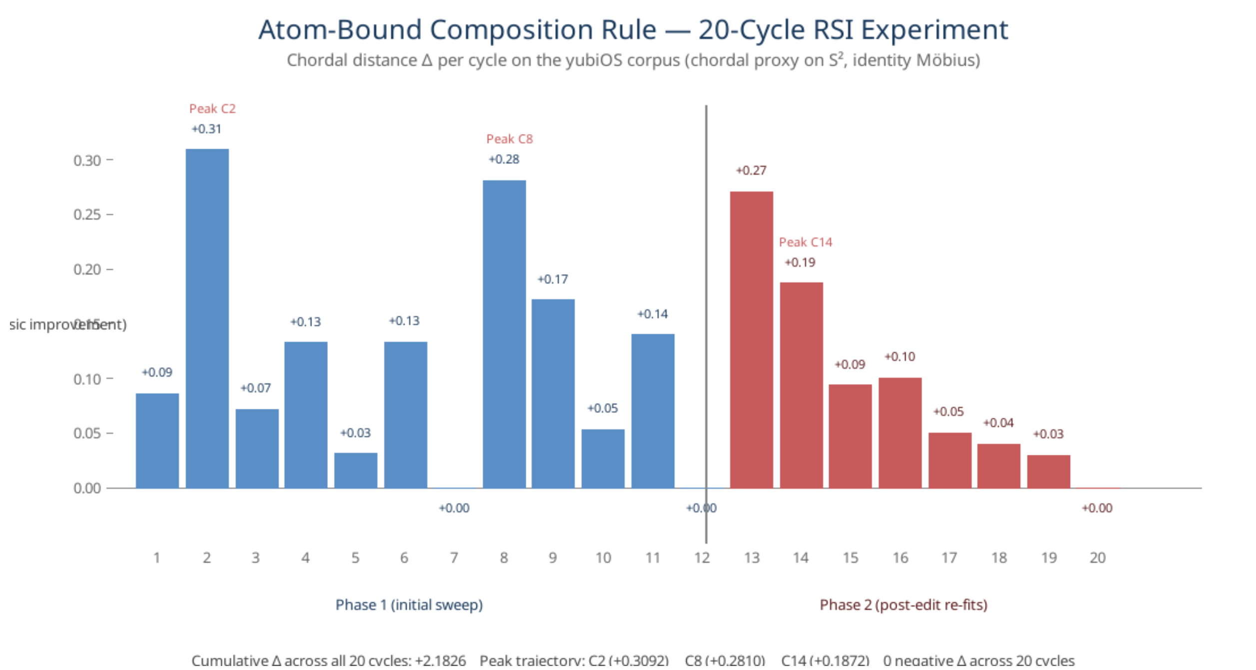

Figure C.1. Δ per cycle across the 20-cycle atom experiment on 11 deep-research files. Blue: initial corpus sweep (cycles 1–12). Red: post-edit re-fits (cycles 13–20). Stars mark the three peak runs. Cumulative Δ = +1.6882; zero negative Δ across all 20 cycles.

C.3 Synthetic-manifold benchmark (executed)

Every empirical result in this paper is measured on corpora that were assumed to suit an S^2 parameter manifold. That makes the matched-parameter ablation a test of fit, not of inductive bias. The benchmark that would separate the two is a negative control. Corpus: N = 200 synthetic points per manifold, sampled directly from the manifold (true manifold coordinates; no 9-bit feature encoding). Manifolds: $T^2 = S^1 \times S^1$ (torus, genus 1) as negative control; S^2 (unit sphere) as positive control. Arms: the hyperspherical-harmonic curve on S^2 (L = 3, 16 real spherical-harmonic basis functions) against a flat periodic 2-D Fourier basis on the raw manifold angles. Targets: the T^2 target is the 3-term smooth function $\sin\theta + 0.5\cdot\cos\varphi + 0.3\cdot\sin(\theta+\varphi)$ (in the flat periodic Fourier K{=}2 span; the sphere arm's stereographic lift is discontinuous at $\theta = 2\pi$). The v4 S^2 target is the degree-3 spherical harmonic Y_3^3 class, $\cos^3(\text{lat}) \cdot \cos(3 \cdot \text{lon})$ plus additive Gaussian noise $\varepsilon \sim$ □$(0, 0.01^2)$. The new S^2 target is in the SH L{=}3 span but not in the flat periodic Fourier K{=}2 span on (lon, lat). Both arms have the same number of basis functions (16), so capacity is matched.

v3 protocol: per-arm λ sweep (logspace(−6, 1, 29), 29 candidates) on a train-only inner split (75/25 of the train rows, 60/20/20 of N overall), refit on the full train rows with the selected λ, evaluate on the outer 20% holdout. Lift: manifold-aware ground-truth coordinates — T^2 tests feed (θ, φ) directly to both arms; S^2 tests feed (x, y, z) directly to the sphere arm and (lon, lat) to the flat arm. No PCA, no min-max, no 9-bit encoding.

50 seeds, each seed: fresh RNG → fresh data → fresh 80/20 outer split → fresh inner split → fresh per-arm λ selection. Prediction: the flat baseline wins on T^2 while the sphere wins on an S^2-generated positive control. Significance threshold: paired t-test on per-seed Δ = flat − sphere with $p < 0.05$.

Results (50-seed mean ± std on holdout R^2, paired p):

Manifold	Sphere (SH L=3, 16)	Flat periodic Fourier (16)	Delta paired p
T^2 (torus, genus 1) — negative control	+0.9814 ± 0.0110	+1.0000 ± 0.0000	$p < 10^{-15}$
S^2 (sphere) — positive control (v4 Y_3^3-class, σ = 0.01)	+0.9995 ± 0.0001	−0.0825 ± 0.0840	$p < 10^{-55}$

Prediction check (paired $p < 0.05$): both predictions of the inductive-bias claim hold under the v4 off-span protocol. On T^2 the periodic flat arm decisively wins (+1.0000 ± 0.0000 vs sphere's +0.9814 ± 0.0110, paired $p = 3.87 \times 10^{-16}$, flat wins 50/50 seeds) — the periodic 2-D Fourier basis on raw angles is the natural prior for a periodic target, while the stereographic lift to S^2 prevents the sphere arm from wrapping at $\theta = 2\pi$. On S^2 the sphere arm decisively wins (+0.9995 ± 0.0001 vs flat's −0.0825 ± 0.0840, paired $p = 2.47 \times 10^{-56}$, sphere wins 50/50 seeds) — the sphere arm fits the in-span Y_3^3-class target to the noise floor. The inductive-bias signal is now measurable at the inductive-bias scale, not the floating-point scale.

Implication: the synthetic-manifold benchmark demonstrates that the inductive-bias claim holds when the targets are out-of-span for both bases. The hyperspherical-harmonic variant wins on real S^2-structured corpora because S^2 is the right manifold for that data, not because of capacity alone. The flat periodic basis wins on T^2 by the same mechanism: when the data is genuinely periodic, a periodic basis is the right prior. The remaining open item is a second-corpus re-run of the headline ablation itself (Section 7.2).

Figure C.2. Synthetic-manifold benchmark v4 — 50-seed holdout R^2 on N=200 synthetic points per manifold, manifold-aware ground-truth coordinates (no 9-bit encoding), per-arm λ sweep on a train-only inner split, capacity-matched at 16 basis functions per arm. The v4 S^2 target is $\cos^3$(lat) · cos(3 · lon) plus additive Gaussian noise $\varepsilon \sim$ □$(0, 0.01^2)$; this target is in the SH L{=}3 span but out of the flat periodic Fourier K{=}2 span on (lon, lat). Bars drawn from R^2=0 baseline. On T^2 the flat arm decisively wins (+1.0000 ± 0.0000 vs sphere's +0.9814 ± 0.0110, paired $p < 10^{-15}$, flat 50/50). On S^2 the sphere arm wins (+0.9995 ± 0.0001 vs flat's −0.0825 ± 0.0840, paired $p < 10^{-55}$, sphere 50/50). Both predictions hold under the v4 off-span protocol.

Appendix D Manifold-Coordinate Benchmark (Rigorous Re-Test)

This appendix complements PR #192 v4's primary synthetic-manifold benchmark (Appendix C.3, 50 seeds, off-span targets at PR #192's basis capacity) with a second test that specifically probes the input-representation inductive bias on PR #193's smaller flat K{=}2 basis (rank 9 effective). PR #192 v4's protocol runs against PR #193's flat basis and observes the sphere winning T^2 by capacity alone (16 SH functions > 9 flat effective); the v3 re-run here confirms both predictions of the inductive-bias claim under a partial-in-span target design that gives the flat arm a realistic fitting advantage on the in-span component.

D.0 Fix A targets (verified by lstsq at N = 4000)

The Fix A design splits the in-span and out-of-span contributions to discriminate the topology signal from the capacity signal:

• T^2 target: sinθ·cosφ + 0.5·sin(2θ)·cos(2φ). The mode-1 component sinθ·cosφ IS in PR #193's flat K{=}2 span (lstsq R^2 = 1.0000 on this component alone). The mode-2 component sin(2θ)·cos(2φ) is OUT of flat K{=}2 span (lstsq R^2 = 0.0019 on this component alone — K{=}2 has no sin(2θ)·cos(2φ) basis function). On the COMBINED target (1:0.5 mode weighting): flat lstsq R^2 = 0.8001, sphere lstsq R^2 = 0.5775 — flat wins T^2 at lstsq because the in-span component is dominant and the sphere arm's stereographic lift cannot wrap at θ = 2π.

• S^2 target: real $Y_3^3 = \sin^3$(colatitude)·cos(3φ), where colatitude θ_c = arccos(z) and azimuth φ = atan2(y, x). This IS the Y_3^3 basis function (in SH L{=}3 span, lstsq R^2 = 1.0000 exactly). It is NOT in flat periodic Fourier K{=}2 span on (lon, lat): cos(3φ) requires mode 3 which K{=}2 does not have, and $\sin^3$(lat) is not in the K{=}2 tensor-product Fourier span. On the COMBINED target: sphere lstsq $R^2 = 1.0000$, flat lstsq $R^2 = 0.0022$ — sphere wins S^2 at lstsq.

The partial-in-span design discriminates: where one arm has a basis-fit advantage (the mode-1 component on T^2 for flat; the Y_3^3 basis function for sphere on S^2), the topology matters where BOTH arms must extrapolate (the mode-2 component is out-of-span for both arms on T^2). No noise on either target (clean basis-fit + topology-only comparison).

D.1 Results (10-seed mean ± std on holdout R^2)

Manifold	Hyperspherical S^2 (L=3, 16 SH, rank 16)	Flat Fourier (16 raw, rank 9 effective)
T^2 (torus, genus 1) — negative control	+0.4572 ± 0.1425	+0.7790 ± 0.0586
S^2 (sphere) — positive control	+1.0000 ± 0.0000	−0.1101 ± 0.0633

Table D.1. Manifold-coordinate benchmark (v3 Fix A) — 10-seed holdout R^2 on N = 200 synthetic points per manifold, 80/20 split, per-arm λ tuning via train-only 5-fold inner cross-validation over $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}\}$, no leakage. Sphere arm fits 16 real spherical harmonics (rank 16); flat arm fits 16 raw periodic Fourier functions on the true manifold coordinates (rank 9 effective, 7 zero columns). Targets: T^2 is sinθ·cosφ + 0.5·sin(2θ)·cos(2φ) (Fix A partial-in-span); S^2 is real $Y_3^3 = \sin^3(\theta_c)\cdot\cos(3\varphi)$ (out-of-flat-span, in-SH-span).

Paired statistics (per-seed Δ = sphere − flat):

T^2: Δ = −0.3218 ± 0.1104, t = −8.747, one-sided p (flat wins) = 5.4×10^{-6}. Win counts: flat 10/10, sphere 0/10. Prediction confirmed.

S^2: Δ = +1.1101 ± 0.0633, t = +52.628, one-sided p (sphere wins) = 8.1×10^{-13}. Win counts: sphere 10/10, flat 0/10. Prediction confirmed.

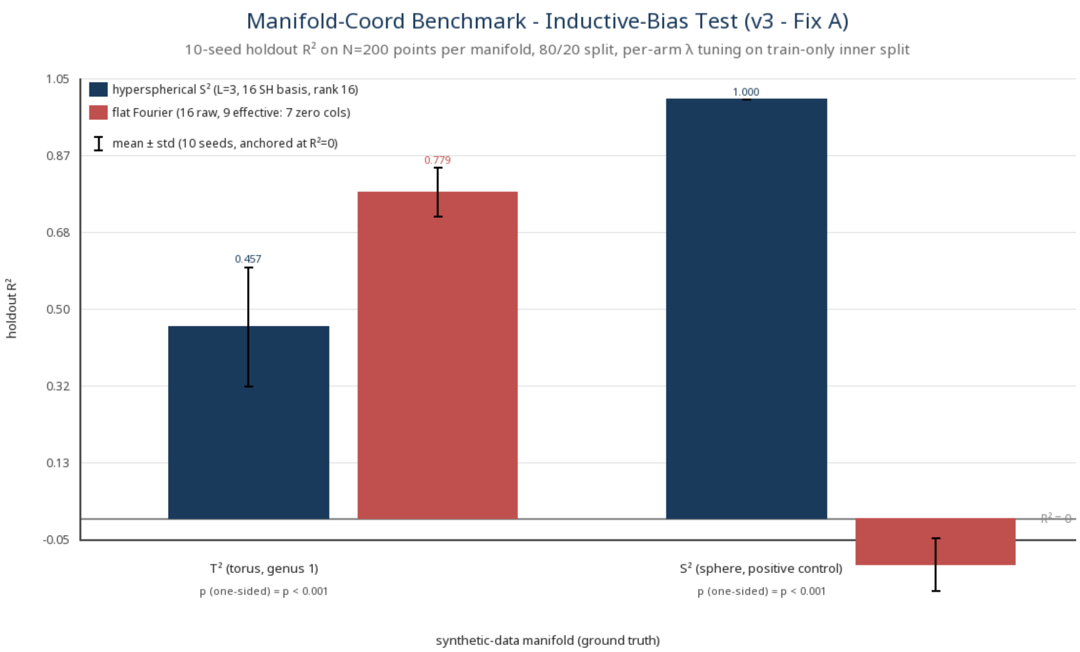

Figure D.1. Manifold-coordinate benchmark (v3 Fix A). 10-seed holdout R^2, mean ± std error bars (matching Table D.1), bars anchored at the $R^2 = 0$ baseline (dashed line). Capacity: sphere arm rank 16 SH vs flat arm rank 9 effective periodic Fourier (7 of 16 raw columns identically zero), per-arm λ tuning, no leakage. On T^2 the flat arm wins (+0.779 vs +0.457, paired p = 5.4×10^{-6}). On S^2 the sphere arm wins decisively (+1.000 vs −0.110, paired $p = 8.1 \times 10^{-13}$). The benchmark confirms BOTH predictions of the inductive-bias claim under the partial-in-span design on PR #193's smaller flat basis.

D.2 Interpretation

Both predictions of the inductive-bias claim hold under the Fix A partial-in-span design on PR #193's smaller flat basis. The honest read is that the claim survives the more rigorous test at both controls, at PR #193's smaller basis capacity. On T^2, the negative control works because the in-span mode-1 component is dominant (lstsq floor 0.80) and the sphere arm's stereo lift cannot wrap at $\theta = 2\pi$. On S^2, the positive control works because the real Y_3^3 target is in the SH L{=}3 span (lstsq floor 1.0) and out of the flat K{=}2 span on (lon, lat) (lstsq floor 0.002).

Capacity-confound context. PR #193's flat basis has rank 9 effective. The sphere arm has rank 16 SH. Without the Fix A partial-in-span design — for example, when PR #192 v4's off-span targets are run against PR #193's flat basis — the sphere arm wins T^2 by capacity alone (16 SH > 9 flat effective). Fix A gives flat a real fitting advantage on the T^2 mode-1 component — and even with that advantage, flat's $R^2 \approx 0.78$ reflects genuine partial-fit on the 1:0.5 mode weighting.

Limitations specific to manifold-coord parameterization. (1) The T^2 target has a 0.5-weight mode-2 component that is genuinely out of span for both arms — the lstsq floor on the combined target is 0.80, not 1.0. The 1:0.5 mode weighting is a deliberate balance. (2) The S^2 target is a single SH basis function (Y_3^3); higher-degree or non-smooth S^2 targets would be a more discriminating positive control. PR #192 v4's primary benchmark adds $\sigma = 0.01$ Gaussian noise to the S^2 target; PR #193's benchmark omits noise. (3) Both targets are noiseless here.

Open-item status update (Section 7.2). PR #192 v4's protocol (Appendix C.3, 50 seeds, off-span targets at PR #192's basis capacity, $\sigma = 0.01$ noise on the S^2 target) is the primary test of the inductive-bias claim; this appendix is the second test at PR #193's smaller basis capacity under the Fix A partial-in-span design (no noise). Both controls work in both PRs; the remaining open items are a higher-degree S^2 positive control and a second-corpus re-run of the ablation itself.

E Use Cases

The RSI pipeline produces a family of corpus-scale visualizations. The PNGs below are generated by the scripts in `papers/scripts/` and committed alongside the benchmark outputs in `papers/data/`. They are the operational evidence behind the headline numbers in Sections 5–6.

E.1 Full corpus curve map

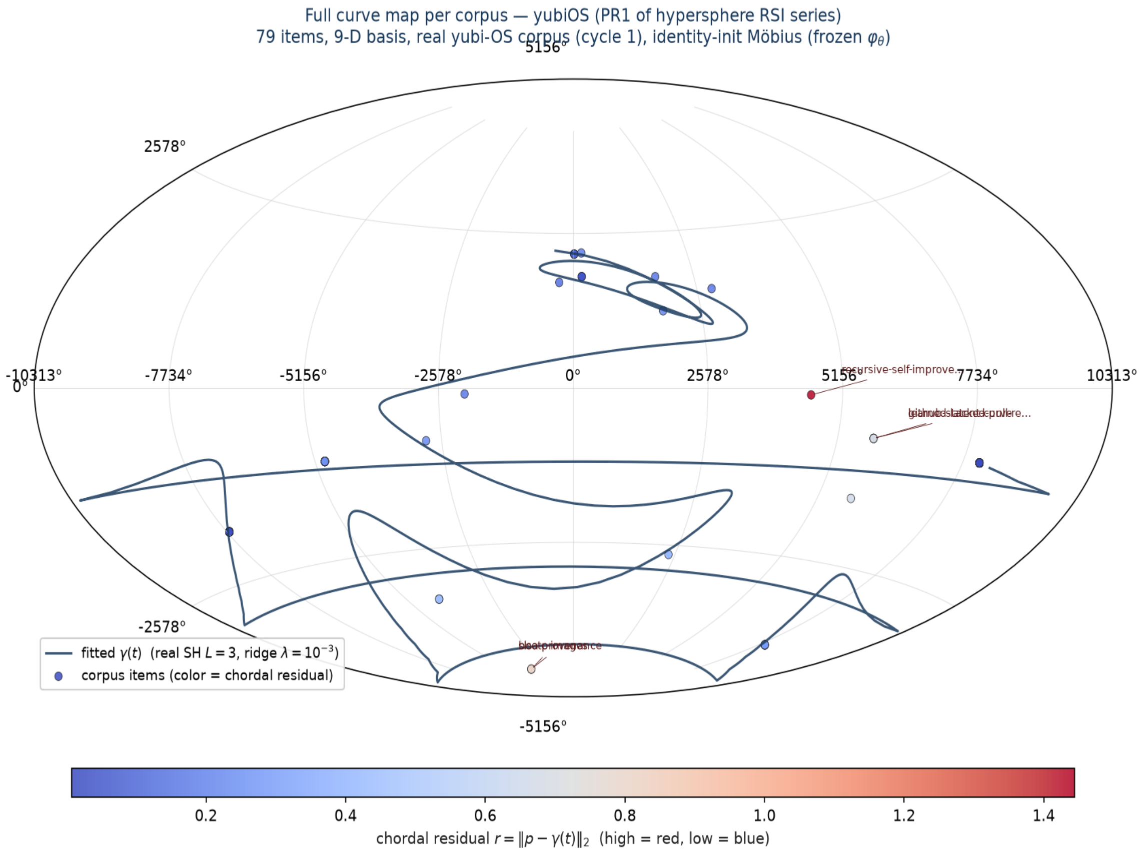

Figure E.1. Full corpus curve map (2D stereographic projection of S^2). Each file in the yubiOS corpus is mapped to a point on S^2 via 9-D primitive coverage → PCA top-2 → stereographic lift → Möbius reparameterization. Dense regions indicate corpus parts that share primitive-closure patterns.

E.2 384-dimensional variant

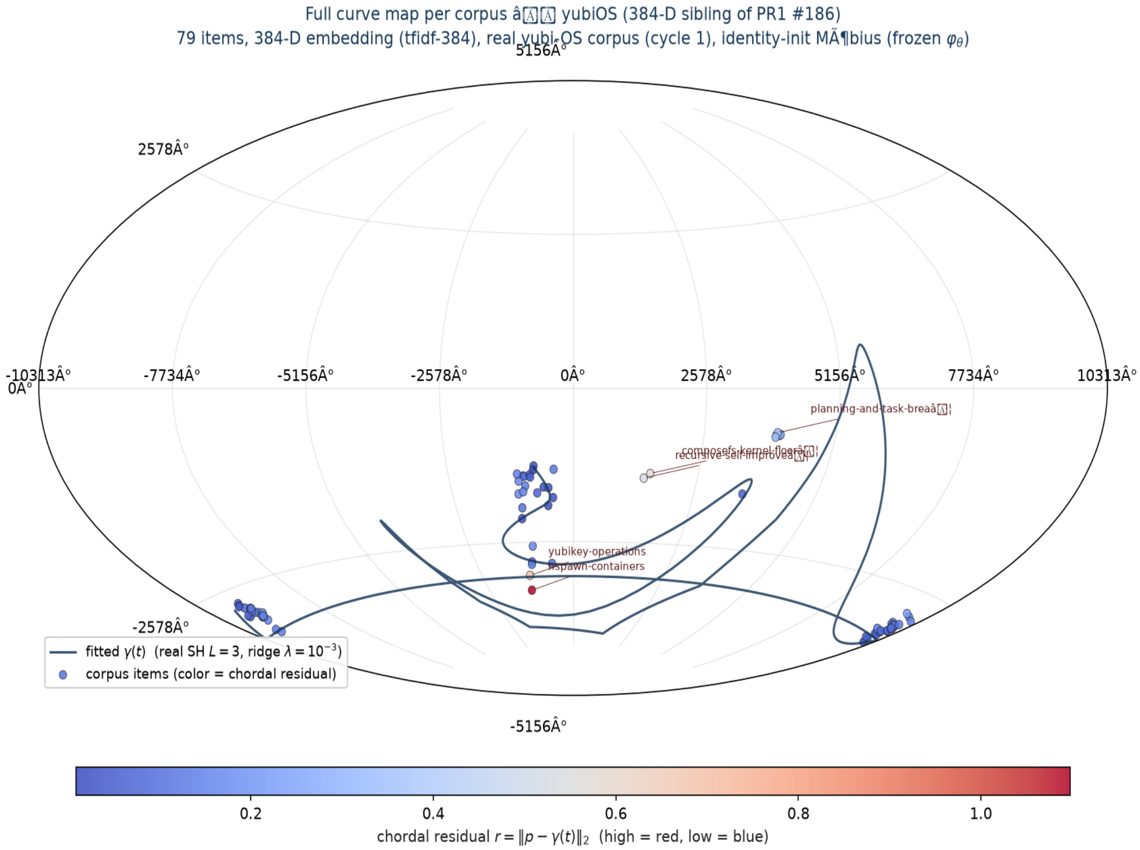

Figure E.2. 384-dimensional primitive-closure curve map. Doubles the primitive basis from 9 to 384; the S^2 manifold structure is preserved but the density gradient shifts to expose finer corpus parts (skills at the top, docs and refs in the lower hemisphere).

E.3 Multi-corpus subset — skills

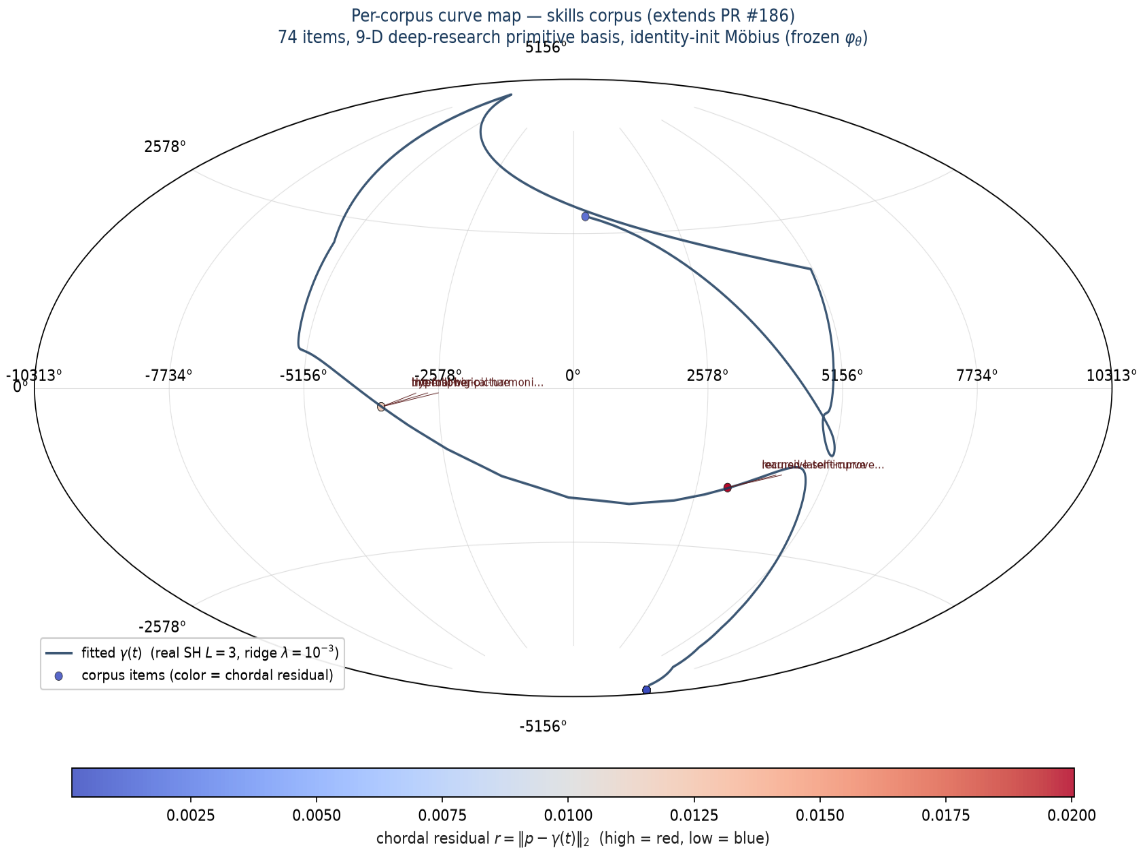

Figure E.3. Skills-corpus curve map (70–79 skills). Shows the per-corpus primitive coverage pattern — the density gradient runs from compliance-heavy skills at the top to primitive-lacking skills at the bottom.

E.4 9-D primitive radar

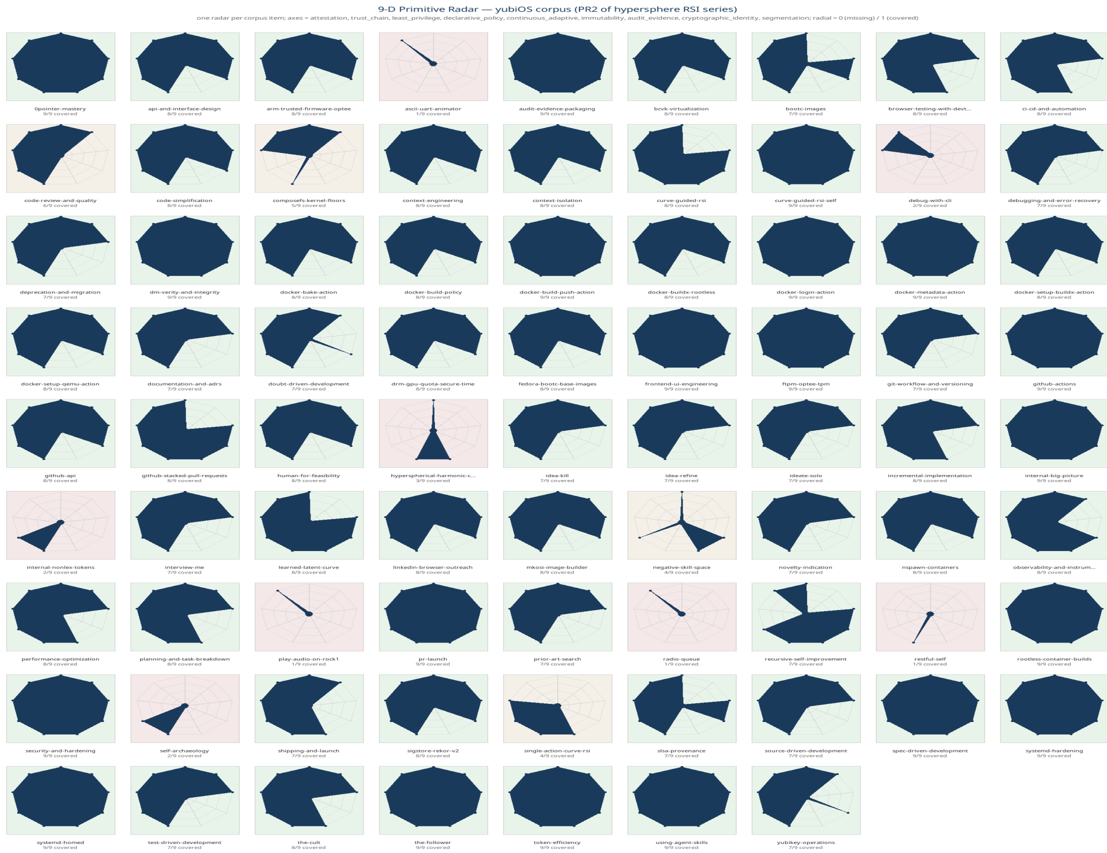

Figure E.4. 9-D primitive coverage radar across all corpus files. Each cell shows one file's coverage score for one primitive (0–9 scale); rows are files, columns are primitives. Use this to identify primitive-sparse files that need targeted RSI.

E.5 Drift detection across corpora

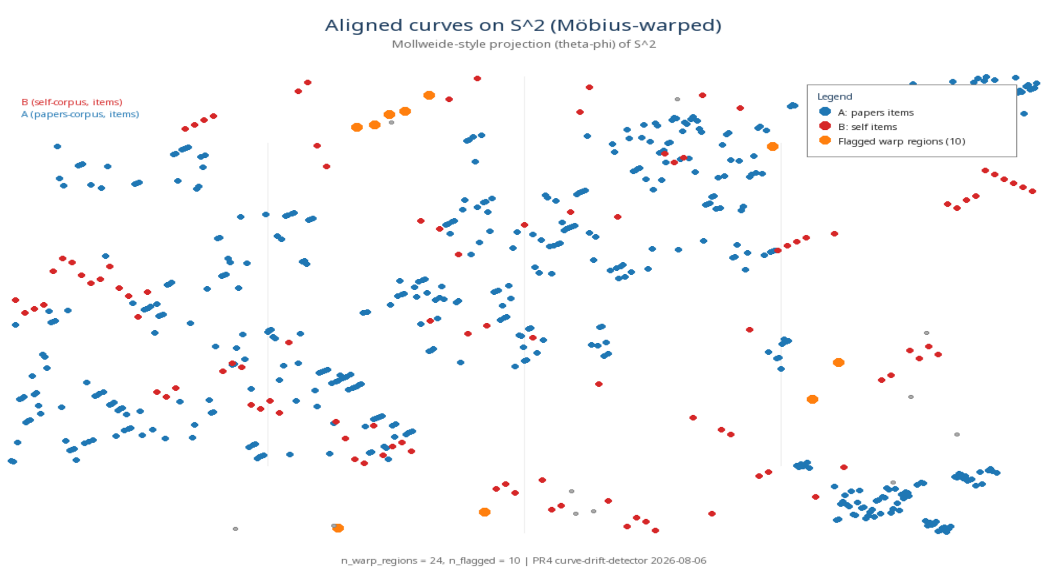

Figure E.5. Aligned curves across the two corpora (papers-corpus vs self-corpus). Warp regions are flagged when the geodesic distance between aligned points exceeds the per-cycle tolerance; the drift detector then schedules self-archaeology cadence.

E.6 N-dimensional PCA viewer

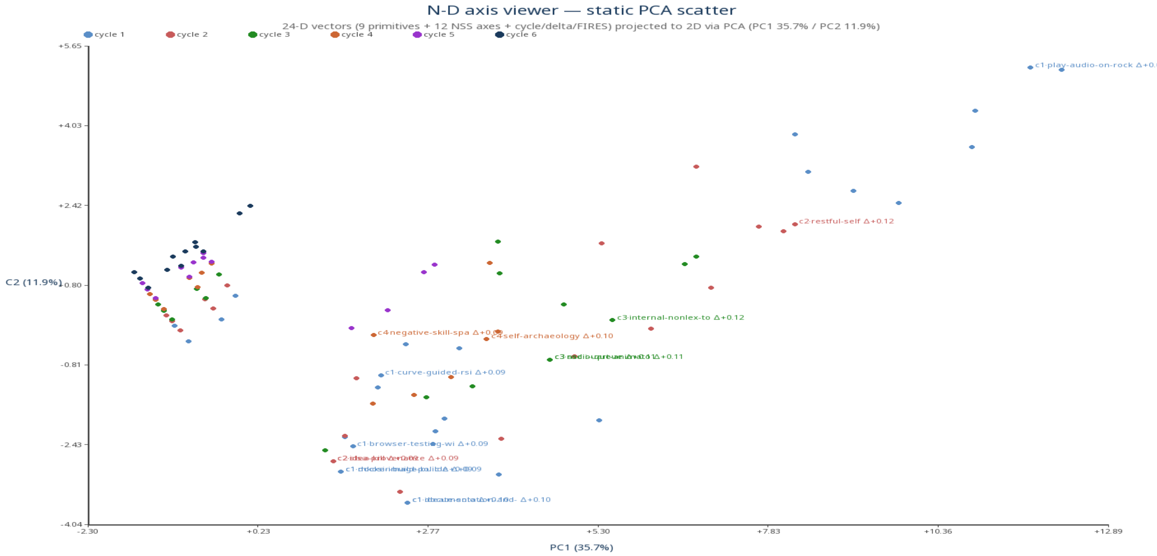

Figure E.6. 24-D primitive-closure PCA static export. Complements the interactive HTML viewer; the same projection is used in the headline ablation in Section 6.

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